



ASTRODONTICS

A Study of Orofacial Effects of
Microgravity and an insight into Aeronautic
dentistry

What is Aeronautic Dentistry?

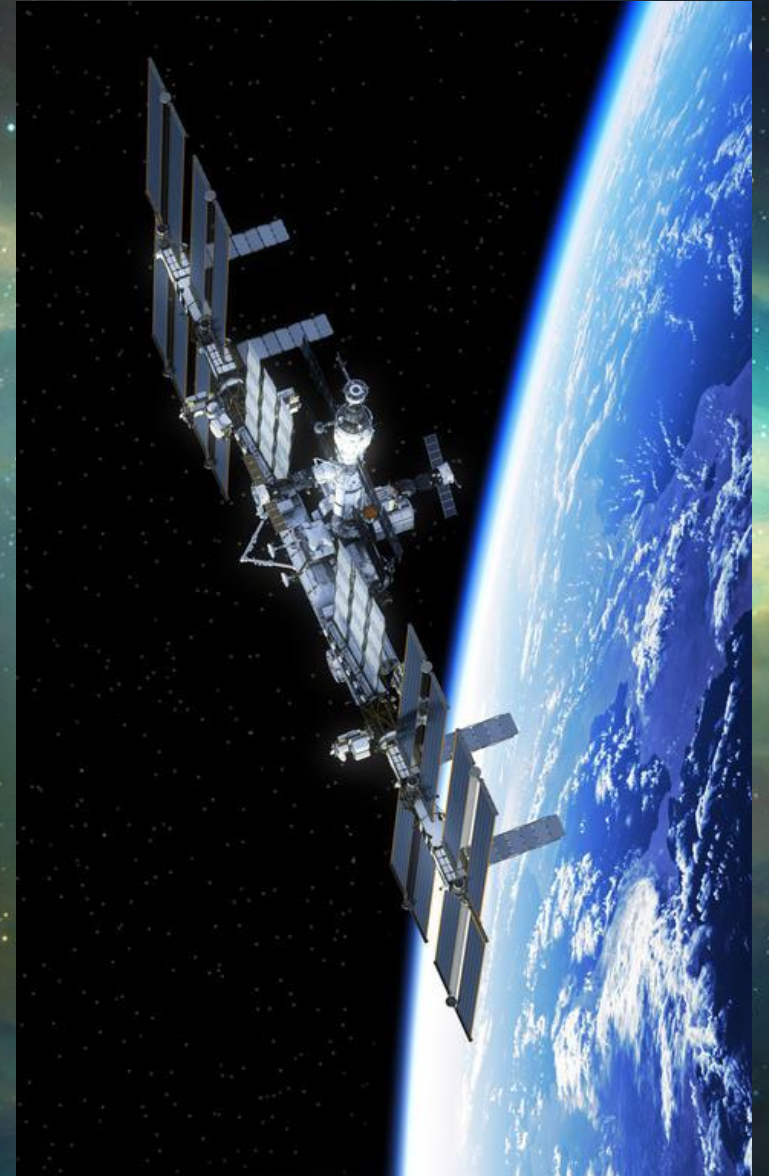


Aeronautic dentistry is the specialized branch of dentistry that deals with the oro-dental factors in aeronautical and extraterrestrial environments.

Aeronautic dentistry also encompasses the diagnosis and treatment modalities that need to be followed in treating diseases in such environments.

WHY IS IT IMPORTANT?

- Increased impetus in Space exploration and space travel.
- Space travel is not a fantasy any more.
- The human body is affected in the most peculiar way in microgravity.
- Future astronauts will require dental care in prolonged journeys.



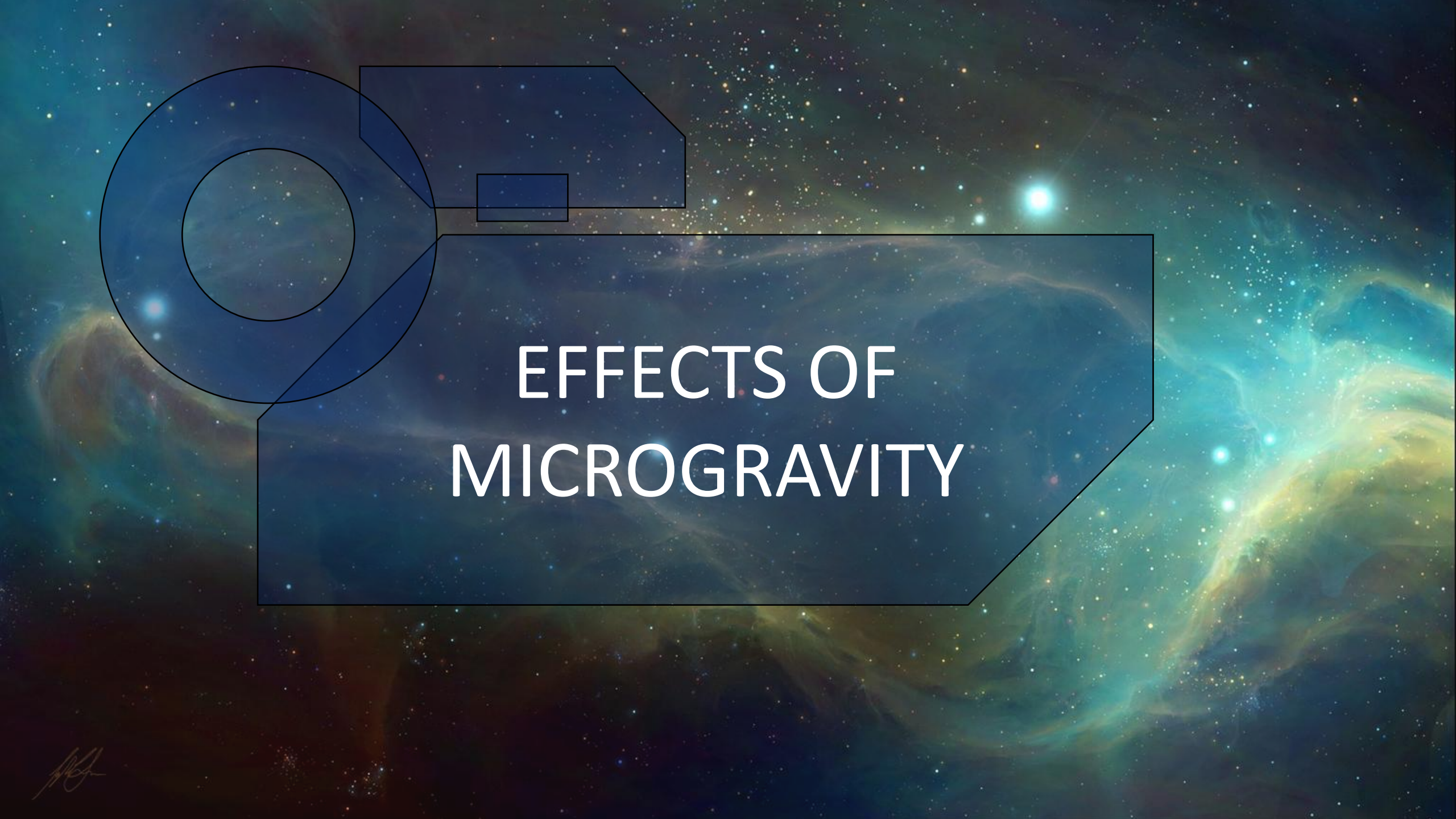
WHAT IS MICROGRAVITY?

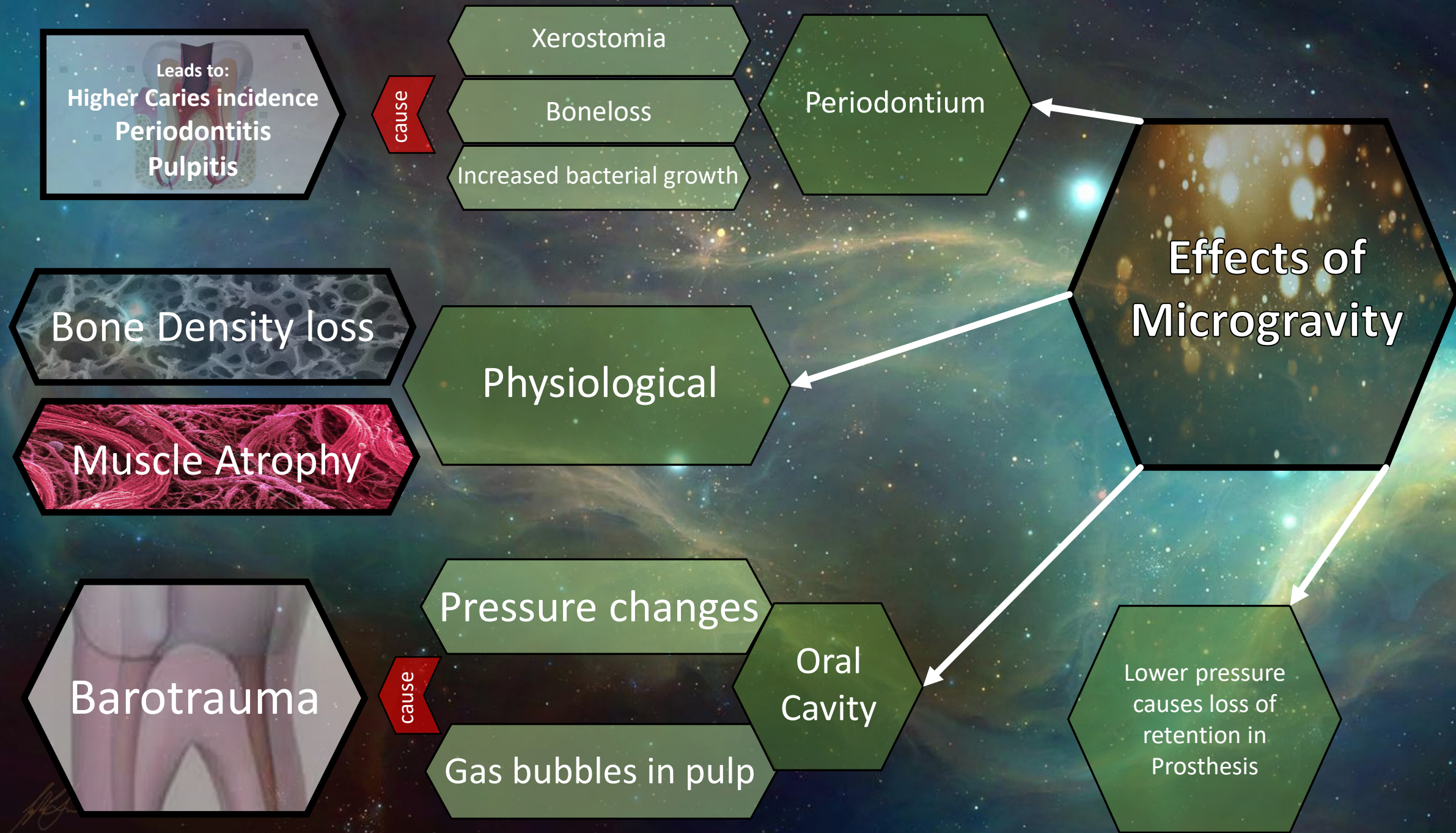
It is a feeling of weightlessness experienced in high altitudes such as in high flying aircrafts and low earth orbits in space due to lower effect of the Earth's gravitational force on objects so far up





EFFECTS OF MICROGRAVITY



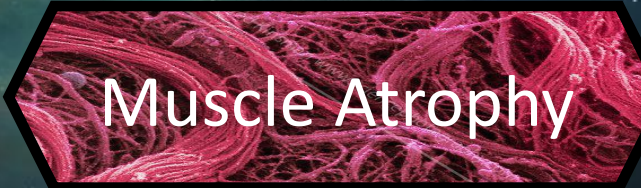




PHYSIOLOGICAL EFFECTS OF MICROGRAVITY



Bone Density loss



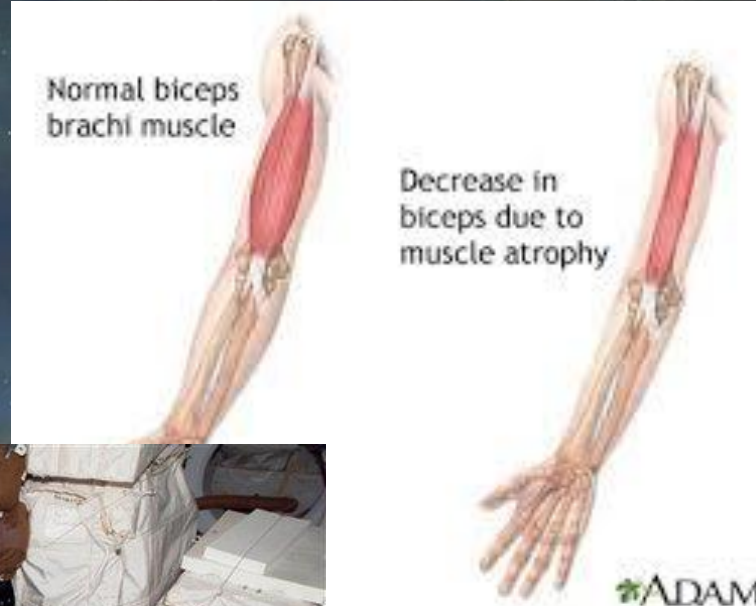
Muscle Atrophy

Bone Density loss



- Reduced stress on bones in microgravity
- Causes loss in the mineral content of bones
- Bones become porous
- Osteoclastin activity increases in microgravity

Muscle Atrophy



- Skeletal Muscle support not needed to maintain posture in microgravity
- Muscle atrophy sets in due to lack of muscle tone

EFFECTS OF MICROGRAVITY ON PERIODONTIUM

Leads to:
Higher Caries incidence
Periodontitis
Pulpitis

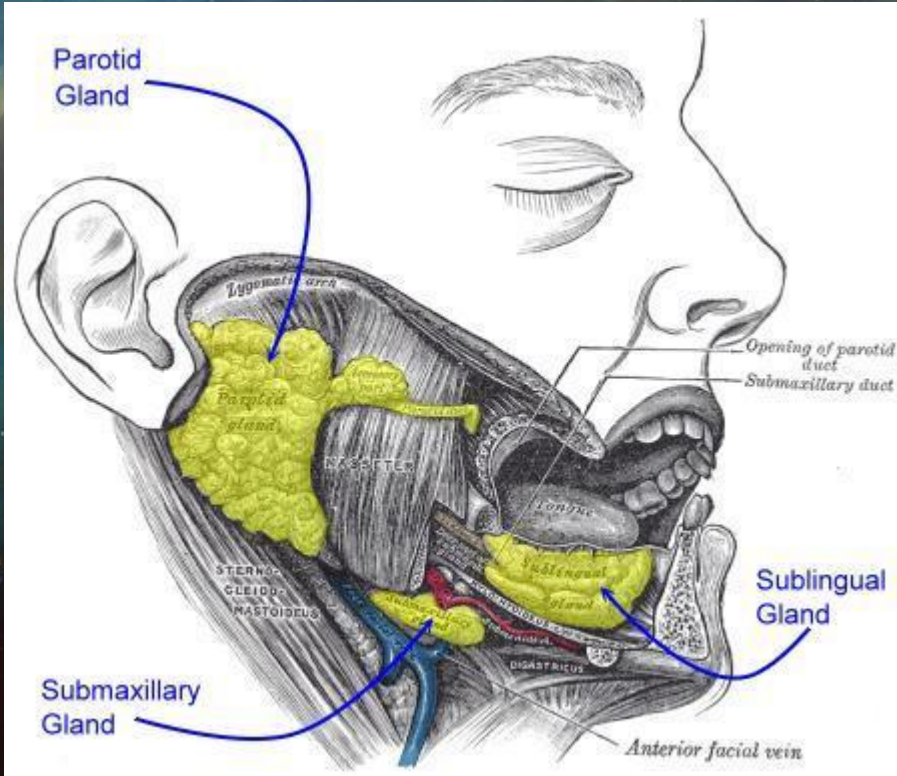
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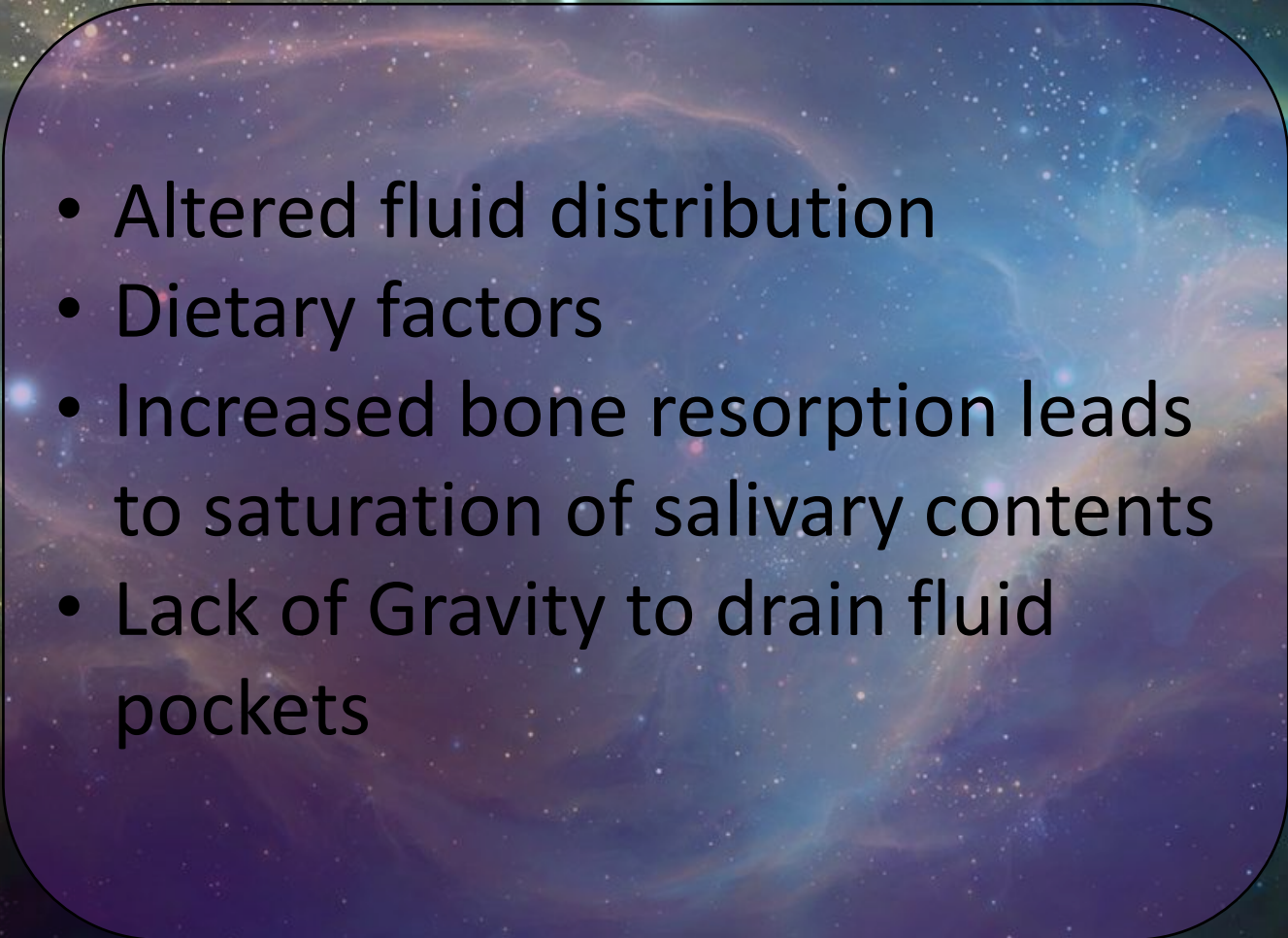
Xerostomia

Bone loss

Increased bacterial growth

Xerostomia



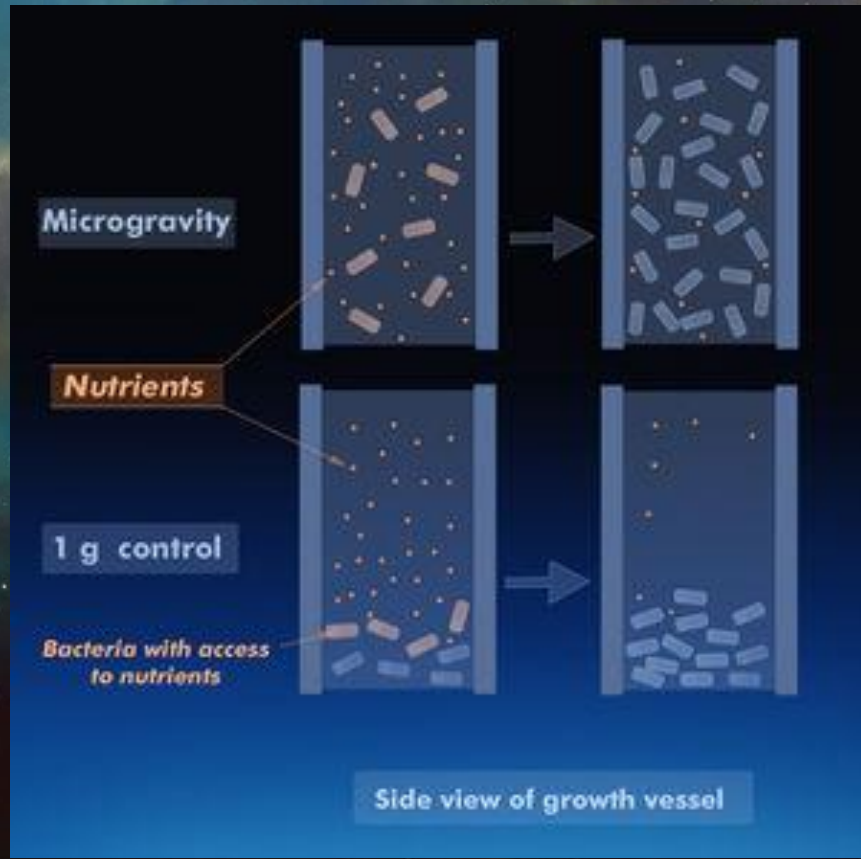
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- Altered fluid distribution
 - Dietary factors
 - Increased bone resorption leads to saturation of salivary contents
 - Lack of Gravity to drain fluid pockets

Bone loss



- Reduced muscle tone
- Reduced pressure
- Altered ion balance
- Changes in diet
- Level of osteocalcin increases in saliva, serum and GCF during microgravity

Increased Bacterial Growth



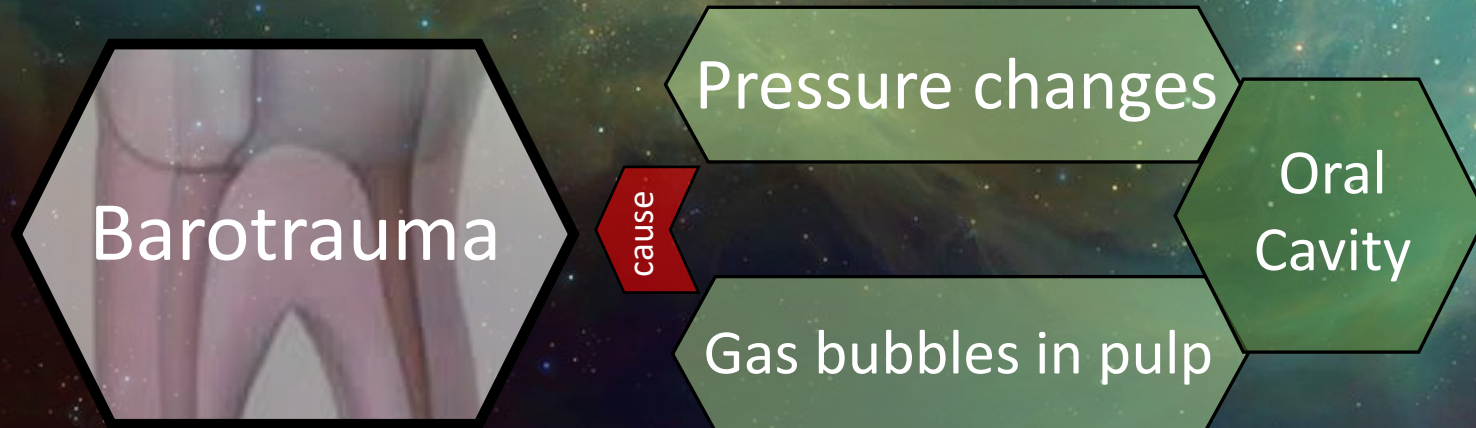
- Isolated environment
- Poorer oral hygiene
- Increased saturation of salivary contents
- Altered gene expression and mass transfer

Leads to:
Higher Caries incidence
Periodontitis
Pulpitis

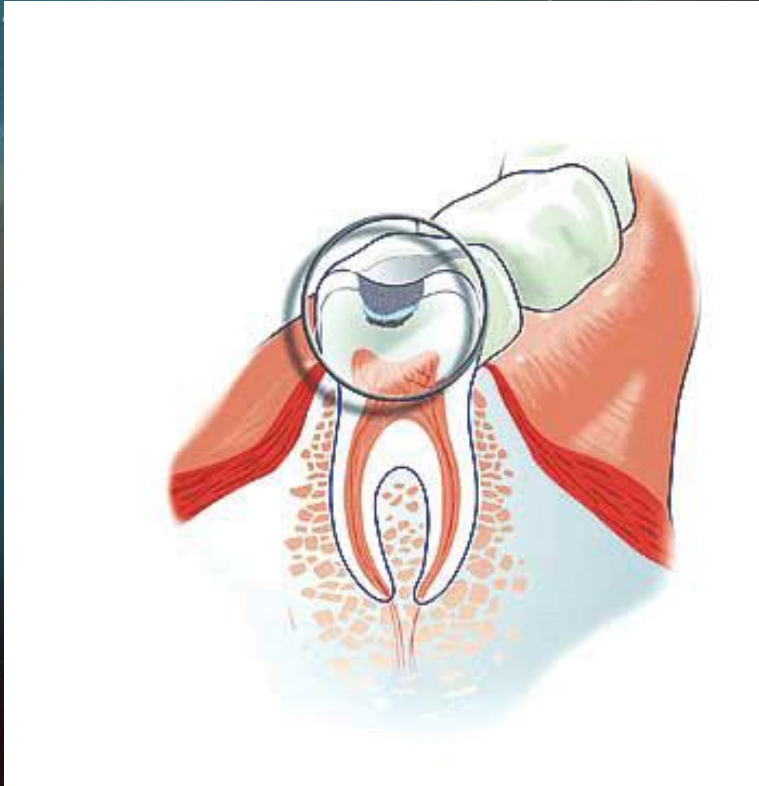


- HIGH BACTERIAL LOAD
- POOR ORAL HYGINE
- BONELOSS
- XEROSTOMIA

EFFECTS OF MICROGRAVITY ON THE ORAL CAVITY

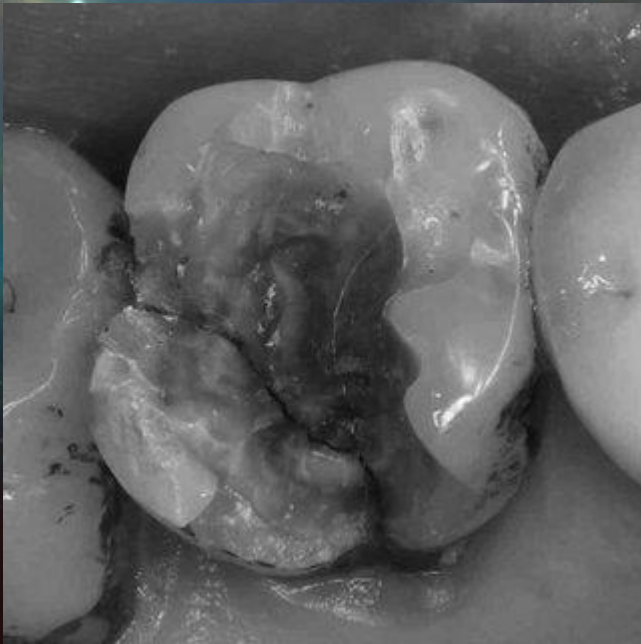


Pressure Changes



- Carious teeth more acutely affected
- Pressure loss during ascent leads to Precipitation of gases in the blood.
- Gas bubbles enter the pulp and cause BARODONTALGIA

Odontocrexis



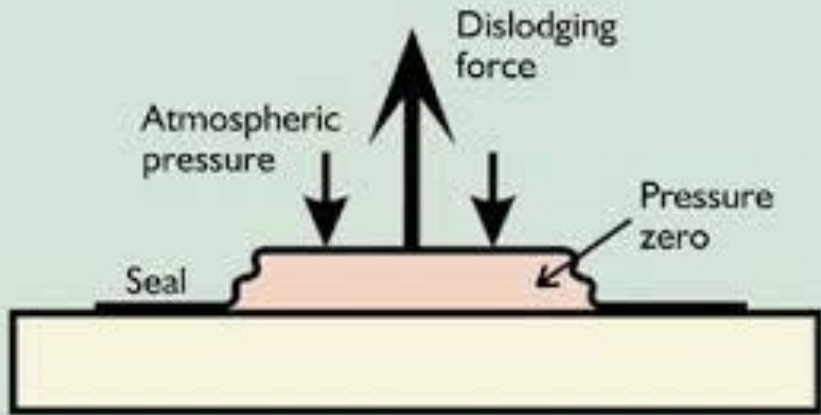
- Term coined by Calder and Ramsey who studied tooth fractures at high altitudes
- Poor restorations may develop leakage or even crack.
- Pressure changes in microtubules may cause fractures of PFM crown



Barotrauma

- Can also occur in divers diving to deep sea levels
- Acute Pain in teeth.
- Fractured restorations or crowns
- Endodontic treatment is indicated but unavailable during missions.

Prosthesis Retention



- Atmospheric pressure is an important factor in retention of removable prosthesis
- Appliances like complete and partial dentures may lose retention in reduced pressure environments.

DENTAL EMERGENCIES IN SPACE

Dental Emergencies in Space

- Two dental emergencies occurred during March 14, 1995, through June 12, 1998.
- One dental emergency occurred during a Mars analogue.
- Two dental emergencies occurred during JBR study of Martian sol study

ORAL CARE FOR ASTROANUTS

- Astronauts must undergo extensive prophylactic evaluation and treatment prior to extensive space travel to avoid dental emergencies.
- Space missions should also be equipped with a specialized dental kit to deal with dental emergencies that may arise on the space station.



ORAL CARE FOR ASTROANUTS

- In space astronauts swallow their toothpaste after brushing to save water and avoid a mess. So the toothpastes need to be non fluoridated (But that increases risk of caries)



NEW FRONTIERS



- With the leaps in space technology by companies like SpaceX, commercial space travel will soon be a reality
- NASA plans to send a manned mission to Mars in a few decades
- Mankind will soon colonize distant planets.

NEED FOR RESEARCH



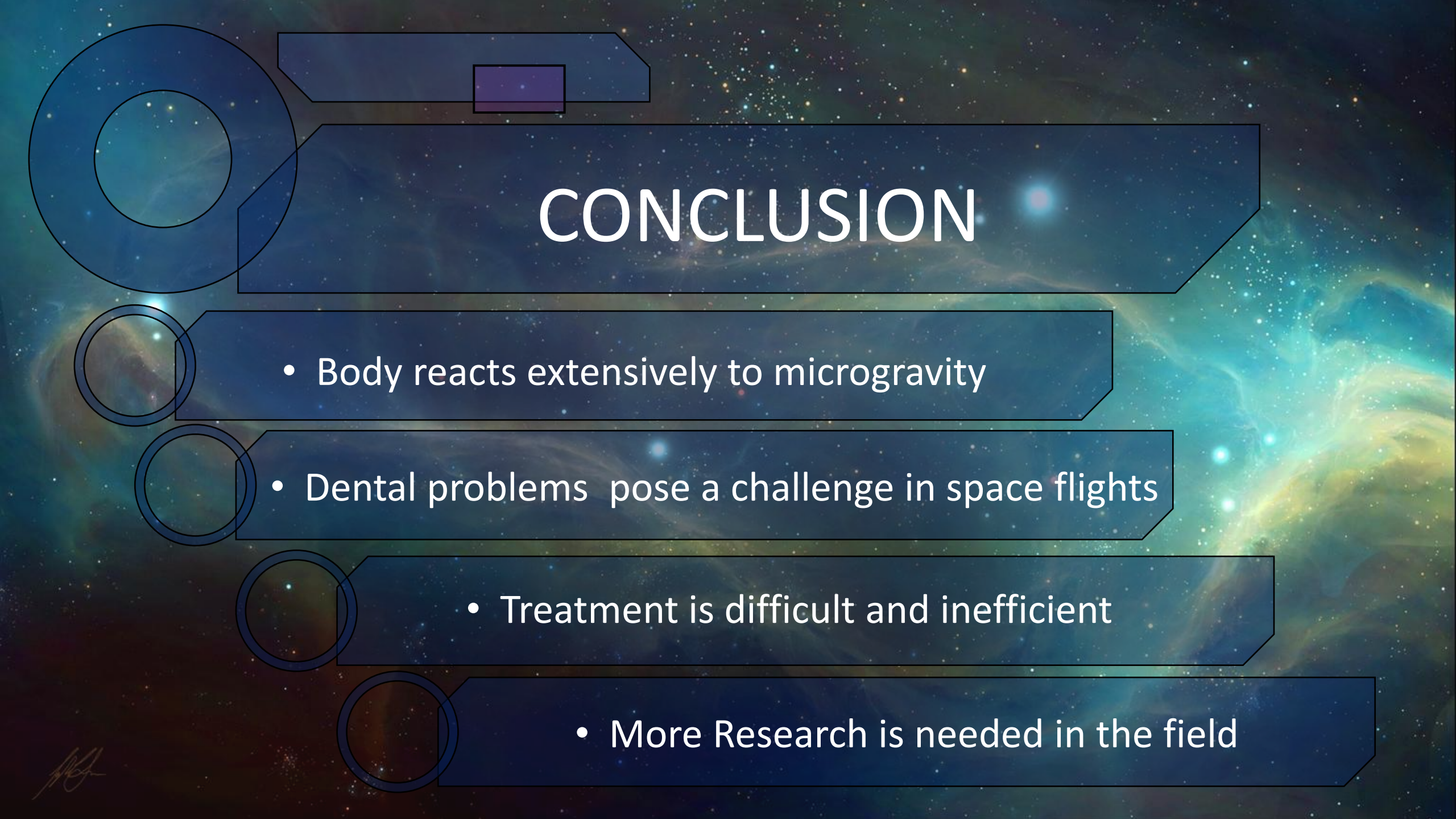
- As more people travel in space need for dental care will increase exponentially
- The technology that is developed may even improve dental practices in conventional treatments



INCLUSION IN CURRICULUM



- Space medicine is an emerging branch in the medical field
- Inclusion of aeronautic dentistry will provide impetus to research, increase awareness and improve practices



CONCLUSION

- Body reacts extensively to microgravity
- Dental problems pose a challenge in space flights
- Treatment is difficult and inefficient
- More Research is needed in the field

References

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- *Assoc. Prof. Dr. Balwant Rai's Colgate article "Space Dentistry: Longer Duration Manned Missions"*
- *B Rai. Aeronautic Dentistry: A New Specialized branch and its Curriculum Guidelines. The Internet Journal of Dental Science. 2006 Volume 5 Number 1.*
- *JPL.NASA.org*
- *<https://www.youtube.com/watch?v=JGJ673cgM4g>*

THANK YOU



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